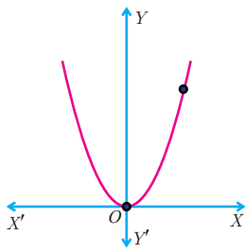
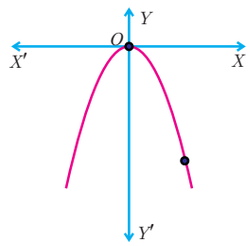
Note

- Modulus function is not a linear function but it is composed of two linear functions x and $-x$.
- Linear functions are always one-one functions and has applications in Cryptography as well as in several branches of Science and Technology.

1.10.3 Quadratic Function

A function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = ax^2 + bx + c, (a \neq 0)$ is called a quadratic function.

Some specific quadratic functions and their graphs

Function, Domain, Range and Definition	Graph
$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2, x \in \mathbb{R}$. $f(x) \in [0, \infty)$	 Fig. 1.47(a)
$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = -x^2, x \in \mathbb{R}$. $f(x) \in (-\infty, 0]$	 Fig. 1.47(b)

The equations of motion of a particle travelling under the influence of gravity is a quadratic function of time. These functions are not one – one. (Why?)

1.10.4 Cubic Function

A function $f : \mathbb{R} \to \mathbb{R}$ defined by

$f(x) = ax^3 + bx^2 + cx + d, (a \neq 0)$ is called a cubic function.

The graph of $f(x) = x^3$ is shown in Fig.1.48.

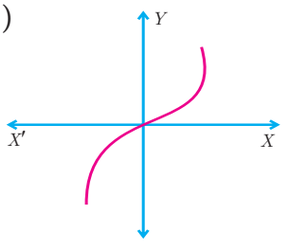


Fig. 1.48

1.10.5 Reciprocal Function

A function $f : \mathbb{R} - \{0\} \to \mathbb{R}$ defined by $f(x) = \frac{1}{x}$

is called a reciprocal function (Fig.1.49).

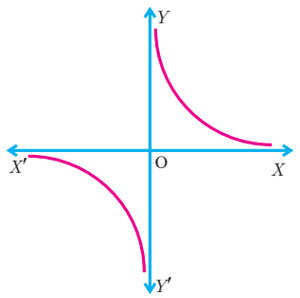


Fig. 1.49

1 Relation and Functions.indd 30 15-12-2020 02:50:39 PM